CHM_159
Worksheet_Chapter_4_Answers

Chemical bonding, chemical nomenclature, molar mass of compounds, mass percent composition, empirical formula, molecular formula and combustion analysis

<u>nomenclature and Lewis dot symbols</u>

I. Complete he following table:

	Cation	Anion	Formula	Name
1	Ba^{2+}	Cl^-	$BaCl_2$	Barium chloride
2	Cr^{3+}	SO_4^{2-}	$Cr_2(SO_4)_3$	Chromium(III) sulfate
3	Ca^{2+}	CO_3^{2-}	$CaCO_3$	Calcium carbonate
4	Mn^{2+}	O^{2-}	MnO	Manganese(II) oxide
5	NH_4^+	SO_4^{2-}	$(NH_4)_2SO_4$	Ammonium sulfate
6	Pb^{4+}	O^{2-}	PbO_2	Lead(IV) oxide
7	Fe^{3+}	CO_3^{2-}	$Fe_2(CO_3)_3$	Iron(III) carbonate
8	K^+	CrO_4^{2-}	K_2CrO_4	Potassium chromate
9	Al^{3+}	PO_4^{3-}	$AlPO_4$	Aluminum phosphate
10	Na^+	$Cr_2O_7^{2-}$	$Na_2Cr_2O_7$	Sodium dichromate
11	Na^+	HCO_3^-	$NaHCO_3$	Sodium bicarbonate
12	Ca^{2+}	Br^-	$CaBr_2$	Calcium bromide
13	Zn^{2+}	ClO_3^-	$Zn(ClO_3)_2$	Zinc chlorate
14	Sn^{2+}	Cl^-	$SnCl_2$	Tin(II) chloride
15	Fe^{2+}	SO_4^{2-}	$FeSO_4$	Iron(II) sulfate
16	NH_4^+	PO_4^{3-}	$(NH_4)_3PO_4$	Ammonium phosphate
17	Hg_2^{2+}	NO_3^-	$Hg_2(NO_3)_2$	Mercury(I) nitrate
18	Cu^{2+}	CH_3COO^-	$Cu(CH_3COO)_2$	Copper(II) acetate

2. Complete the following table:

	Formula	Name of compound
	$ZnSO_4.7H_2O$	Zinc sulfate heptahydrate
	$BaCl_2.\ 2H_2O$	Barium chloride dihydrate
	$CuSO_4.5H_2O$	Copper(ii)sulfate pentahydrate
	$Na_2CO_3.\ 10H_2O$	Sodium carbonate decahydrate
	$MgSO_4.\ 2H_2O$	Magnesium sulfate dihydrate
	$CaSO_4.\ ½H_2O$	Calcium sulfate hemihydrate

3. Write the chemical name and formula for the compounds that contain
   a. five chlorine and one phosphorus atom
   b. one oxygen and two nitrogen atom
   c. two phosphorus and five oxygen atom
   d. one carbon and two oxygen atoms

Ans : PCl_5 phosphorus pentachloride, N_2O dinitrogen monoxide,
P_2O_5 diphosphorus pentoxide, CO_2 carbon dioxide

4. a.Rank the following ionic compounds in terms of decreasing magnitude of lattice energy
(i.e. the absolute value of lattice energy, smallest value last)
NaCl, KCl, CsCl, LiCl

Ans : LiCl > NaCl > KCl >CsCl

b. Rank the following ionic compounds in terms of increasing magnitude of lattice energy
(i.e. the absolute value of lattice energy, largest value last)
KCl, NaCl, $MgCl_2$, MgO, LiCl

Ans : MgO > $MgCl_2$ > LiCl > NaCl > KCl

c. Which ionic compound would you expect to have the higher melting point : NaCl or MgO? Explain your reasoning?

Ans : MgO will have the higher melting point since MgO has a higher magnitude of lattice energy (it is a more exothermic value). Based on Coulomb's law: Lattice energy α (q_1 x q_2)/d where q is the charge on the atoms and 'd' is the distance between them.

grams, mols and atoms

5. a. Calculate the number of C, H, and O atoms in 1.50 g of sucrose, ($C_{12}H_{22}O_{11}$).
Ans : 3.17x10^{22}, H 5.81 x 10^{22}, O 2.90 x 10^{22} atoms
b. How many grams of $Al_2(SO_4)_3$ will contain will contain 1.81 x 10^{24} 'O' atoms
Ans : 85.7 g

mass percent composition

6. Tooth enamel is $Ca_5(PO_4)_3(OH)$. Calculate the percent composition of the elements present.
Ans: Ca 39.9% , P 18.5% O 41.4% H 0.200%

7. Calculate the theoretical percent of water in $CuSO_4.5H_2O$.
Ans : 36.1%

8. Chemical analysis shows that the oxygen-carrying protein hemoglobin is 0.34 percent Fe by mass. The actual molar mass of haemoglobin is about 65000g. How many Fe atoms are in one molecule of haemoglobin.
Ans : 4 Fe atoms

9. Which of the following compounds has the highest percentage of nitrogen by mass : N_2F_4, HNO_3, NH_4NO_2

Ans : NH_4NO_2

10. Analysis of a metal chloride XCl_3 shows that it contains 67.2 percent Cl by mass. Calculate the molar mass of X, and identify the element.
Ans : Cr, MM 52.0 g/mol

11. A mixture of NaBr and Na_2SO_4 contains 29.96 percent Na by mass. Calculate the percent by mass of each compound in the mixture.
Ans : 24.10% NaBr and 75.90% Na_2SO_4 appoximately

12. A mixture of $CuSO_4 \cdot 5H_2O$ and $MgSO_4 \cdot 7H_2O$ is heated until all the water is lost. If 5.020 g of the mixture gives 2.988 g of the anhydrous salts, what is the percent by mass of $CuSO_4 \cdot 5H_2O$ in the mixture?
Ans : 70.70 % approximately

empirical formula and molecular formula

13a. An organic compound consists of 40.00% c, 6.667% H and 53.33% O by mass. The molar mass of the compound is 180. g/mol. What is its molecular formula?
Ans : $C_6H_{12}O_6$

b.Monnosodium glutamate (MSG), a food-flavor enhancer, has been blamed for "Chinese restaurant syndrome," the symptoms of which are headaches and chest

pains. MSG has the following composition by mass: 35.51 percent C, 4.77 percent H, 37.85 percent O, 8.29 percent N, and 13.60 percent Na. What is its molecular formula if its molar mass is about 169 g?

Ans : $C_5H_8O_4NNa$

14. A compound made up of C, H, and Cl contains 55.0 percent Cl by mass. If 9.00 g of the compound contain 4.19×10^{23} H atoms, what is the empirical formula of the compound?

Ans : C_2H_5Cl

15. a. Lactic acid, which consists of C, H, and O, has long been thought to be responsible for muscle soreness following strenuous exercise. Determine the empirical formula of lactic acid given that combustion of a 10.0-g sample produces 14.7 g CO_2 and 6.00 g H_2O. What is the molecular formula if the molecular weight of lactic acid is 90.08 g/mol

Ans : CH_2O, MF $C_3H_6O_3$

b. An 1.68 g of an organic compound containing C, H, and Cl , after undergoing combustion analysis, yields 3.02 g CO_2 and 0.412 g H_2O.. What is the empirical formula of the compound?

Ans : C_3H_2Cl